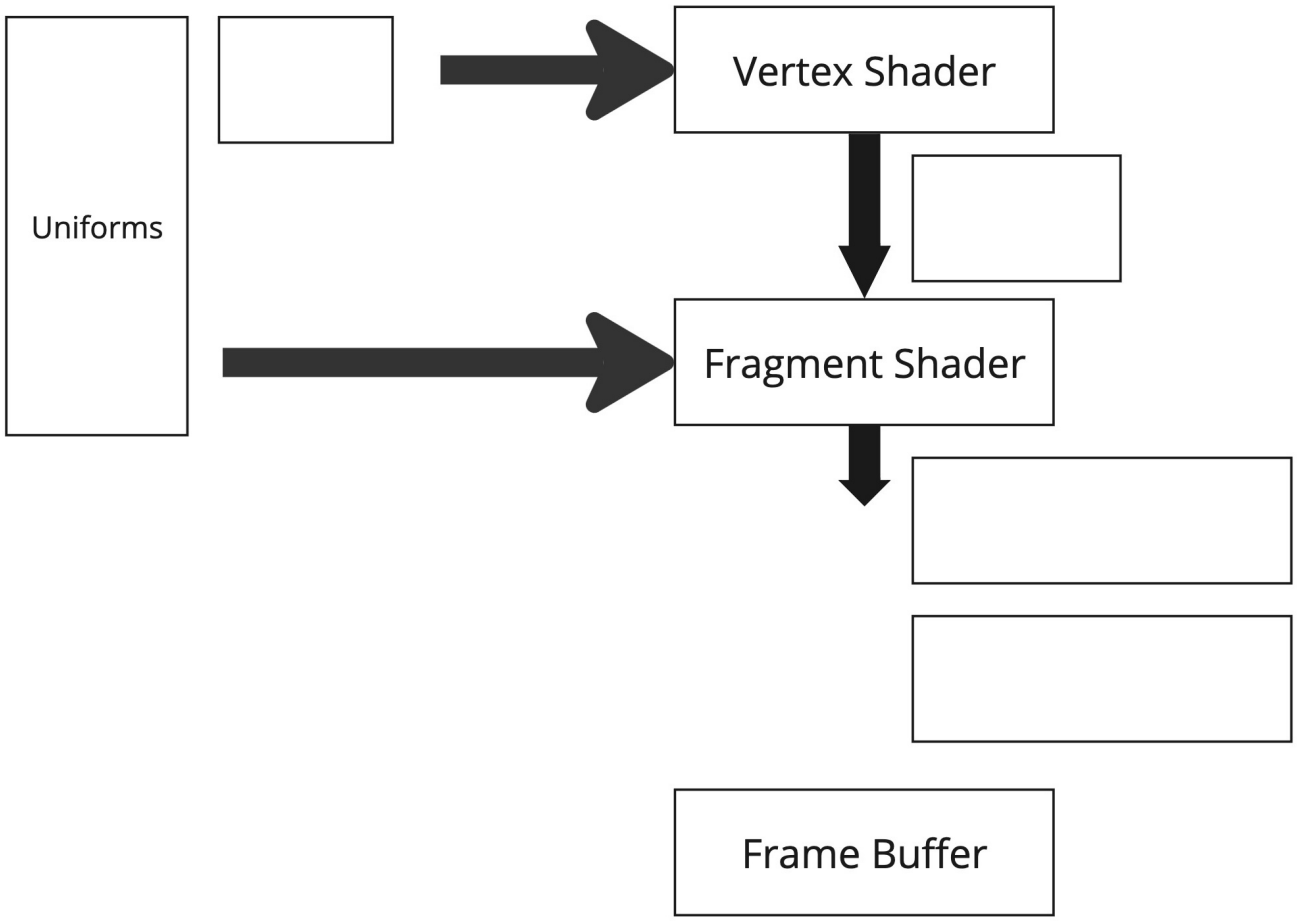
0. Which handles **alpha blending**: the graphics *engine* or the graphics *API* (i.e. OpenGL)?

1. What does `setAnimationFrame` do that a loop and `setInterval` cannot?

2. Please fill in the empty boxes on the following diagram of a **graphics pipeline**.



3. Which of these operates on *interpolated* data (circle one)?

Vertex shader

Fragment shader

4. In addition to the frame buffer, what other buffer is modified by a fragment shader?

5. What is a **draw call**?

6. How are a vertex buffer and an index buffer related?

7. Name three different **render modes**.

8. Give three examples of **affine** transformations.

9. The **depth test** must be configured to properly for WebGL to compare depths. What are 3 different depth test configurations you can use?

10. Given the coordinate (x, y, z), what is the corresponding **homogeneous coordinate** before any other transformations are applied?

11. Why do we need to transform into homogeneous coordinates at all?

12. The **Model Matrix** transforms between what two spaces?

13. The **Projection Matrix** contains two transformations: what are they?

14. What is a **spritesheet**?

15. What is **clip-space** (hint: it is NOT $[0,0]x[1,1]$)?

16. What six planes define a frustum?

17. How do we specify **per vertex** data?

18. If your vertex buffer contains 636 elements, how many elements will your color buffer contain?

19. How do we specify data **between** vertices?

20. True or false: ST-coordinates and UV-coordinates are the same thing (circle one).

True

False

21. Name two configuration options on a **sampler**.

22. What is the difference between a **pixel** and a **texel**?

23. If a texture's dimensions at mip level 0 are `512x512`, what are the dimensions at mip level 1?

24. What is the difference between bilinear and trilinear filtering?

25. True or false: multiplying a UV by a parameter in a fragment shader results in a **dependent texture read** (circle one).

True

False

26. Given the blending function, $b(S, D) = S \cdot sW + D \cdot dW$, what are S and D?

27. This is a five part question: **please answer all five parts**. You have a `100x100` texture filled with values $(x, y, 0)$ at every pixel coordinate (x, y). For example, this means that the value at pixel coordinate $(42, 10)$ is $(42, 10, 0)$. The value at pixel coordinate $(1, 0)$ is $(1, 0, 0)$, and so on.

    What UV coordinate would you need to use to sample the texture at pixel coordinate $(12, 46)$?

    Given the UV coordinate $(0.555, 0.555)$, what value would be returned using the `gl.NEAREST` filter mode?

    Given the UV coordinate $(0.555, 0.555)$, what value would be returned using the `gl.LINEAR` filter mode?

    Given the UV coordinate $(1.1, 1.1)$, what value would be returned using the `gl.NEAREST` filter mode with wrapping `gl.CLAMP`?

    Given the UV coordinate $(1.1, 1.1)$, what value would be returned using the `gl.NEAREST` filter mode with wrapping `gl.REPEAT`?